

Juan David Muñoz Bolaños

Optical Metrology Engineer

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Summary

Physicist and optical engineer finishing a PhD in June 2026, with four years of hands-on lab work building, aligning, and calibrating precision optical measurement systems from scratch. Before that, I spent a year installing and commissioning automated inspection machines at customer sites in industry. I'm comfortable with PC and control technology, used to digging into technical problems systematically, and I enjoy direct contact with the people who use the systems I work on. Based in Tirol, fully authorised to work in Austria, and happy to travel within Austria and nearby countries.

Research/Professional Experience

Jan 2022 to Jun 2026	Medical University of Innsbruck <i>Optical Systems Engineer PhD Researcher</i>	<i>Innsbruck, Austria</i>
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Optomechanical System Design: Built complex laser- and light-based systems from the bare table up, integrating spatial light modulators (SLMs), Texas Instruments DMDs, and galvo scanners for precise light projection.

Automation & Calibration: Developed full C++ and Python control stacks for automated wavefront sensing and real-time phase retrieval, enabling automated calibration routines.

Quality & Testing: Executed structured Design of Experiments (DoE) and optical simulations to verify system performance, applying ISO 9001 and ISO 10110 documentation standards.

Project Ownership: Managed the complete hardware-to-software build of a multiphoton microscope, coordinating with suppliers like Thorlabs and Edmund Optics.

Jan 2020 to Dec 2021	Leibniz Institute of Photonic Technology (IPHT) <i>R&D Research Assistant</i>	<i>Jena, Germany</i>
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Spectroscopy Prototyping: Designed and built a dispersive 1064 nm fiber probe Raman spectrometer, handling optical layout, detector calibration, and system integration.

Data Analysis: Applied supervised machine learning algorithms in Python to classify hyperspectral tissue images with high accuracy.

Jan 2018 to Dec 2019	INTECOL S.A.S. <i>Junior Machine Vision Engineer (Field Service)</i>	<i>Medellín, Colombia</i>
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Machine Vision Pipelines: Engineered custom illumination and Halcon image-processing pipelines for high-speed production lines, improving quality control.

Industrial Automation: Integrated Cognex VisionPro systems for robotic arm guidance, ensuring robust performance in industrial environments.

Technical Competencies

Software & Scripting	Python	C++	LabVIEW/FPGA	MATLAB	Git	ZEMAX
Optical Systems	SLA/DLP Technologies	DMDs	SLMs	Laser Systems	Wavefront Sensing	Illumination Design
Testing & Quality	Design of Experiments (DoE)	ISO 9001 Awareness	ISO 10110	Halcon	OpenCV	

Education

Jan 2022 to Jun 2026	Medical University of Innsbruck Doctor of Philosophy in Optical Metrology	Innsbruck, Austria
Oct 2019 to Sep 2021	Friedrich Schiller Univ. Jena Master of Science in Photonics	Jena, Germany
Aug 2012 to Aug 2018	University of Cauca Bachelor of Science in Physics Engineering	Popayán, Colombia

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

Mountain Sports (Biking, Hiking, Skiing)	Bachata Dancing	Harmonica
Learning German		

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte Medical University of Innsbruck monika.ritsch-marte@i-med.ac.at	PD Dr. Christoph Krafft Leibniz IPHT christoph.krafft@leibniz-ipht.de
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